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Roll No

IT-7005(2)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Artificial Intelligence (AI)

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

iii) Missing data if any assume suitably.

1.
 - a) Describe the water jug problem with their production rule.
 - b) Discuss A* Algorithm and the various observations about algorithm briefly.

2.
 - a) Write the resolution procedure for prepositional logic.
 - b) Consider the following facts and represent them in predicate form.
 - i) There are 500 employees in ABC Company
 - ii) Employees earning more than Rs. 5000 pay tax
 - iii) John is a manager in ABC Company
 - iv) Manager earns Rs. 10,000Convert the facts in predicate form to clauses and then prove by resolution "John pays tax".

3.
 - a) Explain classes and object in frame and give advantages and disadvantages of frame.
 - b) Construct the conceptual dependency representation of the following.
 - i) John gives Mary a book.
 - ii) Ram gave flower to Sita.

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4.
 - a) What is Computer vision? Explain the characteristic of robotics.
 - b) Describe three important properties that neuronal signal functions should possess. Why are these important?
5.
 - a) What is an expert system? Explain all the components of expert system.
 - b) What is back propagation? Explain the back propagation training algorithm with the help of a one hidden layer feed forward network.
6.
 - a) Describe Bayes theorem? Show how it is useful to handle uncertain data?
 - b) Design the heuristic function for the following:
 - i) 8 puzzle problem
 - ii) Travelling salesman problem
7.
 - a) Give an example for which breath first search would work better than depth first search.
 - b) Explain the concept learning using decision trees and neural network approach.
8. Write Short Notes on (Any Three):
 - i) Alpha beta cut off
 - ii) Fuzzy logic
 - iii) Forward and backward reasoning
 - iv) AO* algorithm

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